

Database System Engineering and Distributed Backend Development

Course Code: 25CS13002E

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Query 1: -CREATE DATABASE

```
2520030597-Vidhath  
CREATE DATABASE IF NOT EXISTS bookflow0_db;  
USE bookflow0_db;
```

Output: -

✓ 12	17:07:56	CREATE DATABASE IF NOT EXISTS bookflow0_db	1 row(s) affected	0.015 sec
✓ 13	17:08:04	USE bookflow0_db	0 row(s) affected	0.000 sec

Query 2: -CREATE TABLE

```
CREATE TABLE orders (  
  ord_no INT PRIMARY KEY,  
  purch_amt DECIMAL(10,2),  
  ord_date DATE,  
  customer_id INT,  
  salesman_id INT  
);
```

Output: -

✓ 14	21:36:23	CREATE TABLE orders (ord_no INT PRIMAR...	0 row(s) affected	0.125 sec
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Query 3: - INSERT

```

2520030597-Vidhath
INSERT INTO orders VALUES
(70001,150.50,'2012-10-05',3005,5002),
(70009,270.65,'2012-09-10',3001,5005),
(70002,65.26,'2012-10-05',3002,5001),
(70004,110.50,'2012-08-17',3009,5003),
(70007,948.50,'2012-09-10',3005,5002),
(70005,2400.60,'2012-07-27',3007,5001),
(70008,5760.00,'2012-09-10',3002,5001),
(70010,1983.43,'2012-10-10',3004,5006),
(70003,2480.40,'2012-10-10',3009,5003),
(70012,250.45,'2012-06-27',3008,5002),
(70011,75.29,'2012-08-17',3003,5007),
(70013,3045.60,'2012-04-25',3002,5001);

```

Output: -

✓ 15 21:37:55 INSERT INTO orders VALUES (70001,150.50... 12 row(s) affected Records: 12 Duplicates: 0 Wa... 0.047 sec

Query 4: - WHERE CLAUSE

```

2520030597-Vidhath
SELECT *
FROM orders
WHERE purch_amt > 2000;

```

Output: -

✓ 16 21:40:16 SELECT * FROM orders WHERE purch_amt > ... 4 row(s) returned 0.000 sec / 0.000 sec

Result Grid		Filter Rows:	Export: Wrap Cell Content		
	ord_no	purch_amt	ord_date	customer_id	salesman_id
▶	70003	2480.40	2012-10-10	3009	5003
	70005	2400.60	2012-07-27	3007	5001
	70008	5760.00	2012-09-10	3002	5001
	70013	3045.60	2012-04-25	3002	5001

Query 5: -WHERE CLAUSE

2520030597-Vidhath

```
SELECT *  
FROM orders  
WHERE ord_date='2012-09-10';
```

Output: -

✓ 17 21:42:11 SELECT * FROM orders WHERE ord_date='20... 3 row(s) returned 0.031 sec / 0.000 sec

Result Grid		Filter Rows:	Export: Wrap Cell Content		
	ord_no	purch_amt	ord_date	customer_id	salesman_id
▶	70007	948.50	2012-09-10	3005	5002
	70008	5760.00	2012-09-10	3002	5001
	70009	270.65	2012-09-10	3001	5005

Query 5: -ORDER BY

2520030597-Vidhath

```
SELECT *  
FROM orders  
ORDER BY purch_amt DESC;
```

Output: -

✓ 18 21:43:46 SELECT * FROM orders ORDER BY purch_amt... 12 row(s) returned 0.000 sec / 0.000 sec

Result Grid		Filter Rows:	Export: Wrap Cell Content		
	ord_no	purch_amt	ord_date	customer_id	salesman_id
▶	70008	5760.00	2012-09-10	3002	5001
	70013	3045.60	2012-04-25	3002	5001
	70003	2480.40	2012-10-10	3009	5003
	70005	2400.60	2012-07-27	3007	5001
	70010	1983.43	2012-10-10	3004	5006
	70007	948.50	2012-09-10	3005	5002
	70009	270.65	2012-09-10	3001	5005
	70012	250.45	2012-06-27	3008	5002
	70001	150.50	2012-10-05	3005	5002
	70004	110.50	2012-08-17	3009	5003
	70011	75.29	2012-08-17	3003	5007
	70002	65.26	2012-10-05	3002	5001

Query 6: -ORDER BY

```
2520030597-Vidhath
SELECT *
FROM orders
ORDER BY ord_date;
```

Output: -

✓ 19 22:55:05	SELECT * FROM orders ORDER BY ord_date ...	12 row(s) returned	0.000 sec / 0.000 sec
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Result Grid		Filter Rows:	Export: Wrap Cell Content		
	ord_no	purch_amt	ord_date	customer_id	salesman_id
▶	70013	3045.60	2012-04-25	3002	5001
	70012	250.45	2012-06-27	3008	5002
	70005	2400.60	2012-07-27	3007	5001
	70004	110.50	2012-08-17	3009	5003
	70011	75.29	2012-08-17	3003	5007
	70007	948.50	2012-09-10	3005	5002
	70008	5760.00	2012-09-10	3002	5001
	70009	270.65	2012-09-10	3001	5005
	70001	150.50	2012-10-05	3005	5002
	70002	65.26	2012-10-05	3002	5001
	70003	2480.40	2012-10-10	3009	5003
	70010	1983.43	2012-10-10	3004	5006

Query 7: -SUM()

```
SELECT SUM(purch_amt) AS total_revenue
FROM orders;
```

Output: -

✓ 20 22:57:35 SELECT SUM(purch_amt) AS total_revenue ... 1 row(s) returned 0.015 sec / 0.000 sec

Result Grid	Filter Rows:	Export: Wrap Cell Content
total_revenue		
▶ 17541.18		

Query 8: -AVG()

```
SELECT AVG(purch_amt) AS average_order
FROM orders;
```

Output: -

✓ 21 22:59:14 SELECT AVG(purch_amt) AS average_order ... 1 row(s) returned 0.000 sec / 0.000 sec

Result Grid	Filter Rows:	Export: Wrap Cell Content
average_order		
▶ 1461.765000		

Query 9: -MAX()

```
2520030597-Vidhath
SELECT MAX(purch_amt) AS highest_order
FROM orders;
```

Output: -

✓ 23 23:01:55 SELECT MAX(purch_amt) AS highest_order ... 1 row(s) returned 0.000 sec / 0.016 sec

Result Grid	Filter Rows:	Export: Wrap Cell Content
highest_order		
▶ 5760.00		

Query 10: - MIN()

```
SELECT MIN(purch_amt) AS lowest_order
FROM orders;
```

Output: -

24 23:04:09 SELECT MIN(purch_amt) AS lowest_order F... 1 row(s) returned 0.000 sec / 0.000 sec

Result Grid		Filter Rows:	Export: Wrap Cell Content
	lowest_order		
▶	65.26		

Query 11: -COUNT()

```
SELECT COUNT(*) AS total_orders
FROM orders;
```

Output: -

25 23:06:00 SELECT COUNT(*) AS total_orders FROM or... 1 row(s) returned 0.031 sec / 0.000 sec

Result Grid		Filter Rows:	Export: Wrap Cell Content
	total_orders		
▶	12		

Query 12: - GROUP BY

```
2520030597-Vidhath
SELECT customer_id,
       SUM(purch_amt) AS total_purchase
FROM orders
GROUP BY customer_id;
```

Output: -

26 23:08:16 SELECT customer_id, SUM(purch_amt) AS t... 8 row(s) returned 0.000 sec / 0.000 sec

Result Grid		Filter Rows:	Export: Wrap Cell Content
	customer_id	total_purchase	
▶	3005	1099.00	
	3002	8870.86	
	3009	2590.90	
	3007	2400.60	
	3001	270.65	
	3004	1983.43	
	3003	75.29	
	3008	250.45	

Query 13: - GROUP BY + MAX()

```
2520030597-Vidhath
SELECT customer_id,
       MAX(purch_amt) AS highest_purchase
FROM orders
GROUP BY customer_id;
```

Output: -

27 23:18:40 SELECT customer_id, MAX(purch_amt) AS h... 8 row(s) returned 0.016 sec / 0.000 sec

Result Grid		Filter Rows:	Export: Wrap Cell Content
	customer_id	highest_purchase	
▶	3005	948.50	
	3002	5760.00	
	3009	2480.40	
	3007	2400.60	
	3001	270.65	
	3004	1983.43	
	3003	75.29	
	3008	250.45	

Query 14: - GROUP BY + HAVING

```
SELECT customer_id,
       COUNT(*) AS total_orders
FROM orders
GROUP BY customer_id
HAVING COUNT(*) > 1;
```

Output: -

28 23:21:25 SELECT customer_id, COUNT(*) AS total_o... 3 row(s) returned 0.000 sec / 0.000 sec

Result Grid		Filter Rows:	Export: Wrap Cell Content
	customer_id	total_orders	
▶	3005	2	
	3002	3	
	3009	2	

Query 15: - GROUP BY + HAVING + ORDER BY

```
SELECT customer_id,
       SUM(purch_amt) AS total_purchase
FROM orders
GROUP BY customer_id
HAVING SUM(purch_amt) > 1000
ORDER BY total_purchase DESC;
```

Output: -

29 23:23:45 SELECT customer_id, SUM(purch_amt) AS t... 5 row(s) returned 0.000 sec / 0.000 sec

Result Grid			Filter Rows:	Export: Wrap Cell Content
	customer_id	total_purchase		
▶	3002	8870.86		
	3009	2590.90		
	3007	2400.60		
	3004	1983.43		
	3005	1099.00		

Query 16: - BETWEEN + HAVING

```
2520030597-Vidhath
SELECT customer_id,
       MAX(purch_amt) AS max_purchase
FROM orders
GROUP BY customer_id
HAVING MAX(purch_amt) BETWEEN 2000 AND 6000;
```

Output: -

30 23:25:35 SELECT customer_id, MAX(purch_amt) AS m... 3 row(s) returned 0.000 sec / 0.000 sec

Result Grid			Filter Rows:	Export: Wrap Cell Content
	customer_id	max_purchase		
▶	3002	5760.00		
	3009	2480.40		
	3007	2400.60		

Query 17: - MAX + GROUP BY + HAVING


```
SELECT ord_date,  
       MAX(purch_amt) AS highest_purchase  
FROM orders  
GROUP BY ord_date  
HAVING MAX(purch_amt) > 2000;
```

Output: -

31 23:27:25 SELECT ord_date, MAX(purch_amt) AS high... 4 row(s) returned 0.016 sec / 0.000 sec

Result Grid		Filter Rows:	Export Wrap Cell Content
	ord_date	highest_purchase	
▶	2012-10-10	2480.40	
	2012-07-27	2400.60	
	2012-09-10	5760.00	
	2012-04-25	3045.60	